

Figures for manuscript

Overview

There are 11 different mixtures of alignments: whole genome, 10% wgs-90% pol, 20% wgs-80% pol, and so on up until 100% pol. For the mixtures i.e. 10% wgs-90% pol up to 100% pol, we created 1000 samples of each by using different seeds to randomize the rows (sequences) and start the masking procedure. Then we inferred trees with iqtree, leading to 1000 sets of tree mixtures. There are 1196 tips on these trees, which comprise all subtype B.

Running this notebook should reproduce all figures in the manuscript as long as the appropriate packages are installed. Some code for additional results and for different sections are included as well.

Main findings

- Distance between mixture trees and wgs tree remains around the same up to **wgs50**, then steadily decreases (i.e. mixture gets closer to wgs tree)
- Congruence between bootstrap values on mixture trees and wgs tree is minimal at **wgs60**.
- In terms of congruence between **pol** and the mixtures with clusters identified with **pol**, there is a dip to **wgs60**, and then congruence increases again. (Figures @ref(fig:lineplot-clusterpicker) and @ref(fig:heatmap-clusterpicker-bootstrap))

Setup code chunks

We have a series of setup code chunks:

- **setup:** Loads packages, all tree bootstraps (the most compute-intensive part so was saved as an object), alignment distances, manuscript specific code
- **loadtrees:** Loads all trees, including maskgp120, samples, fixed parameter trees
- **cluster_picker_load, alignment_load, get_bootstrap:** not run, creates objects loaded in **setup** and **loadtrees**
- **make_data_frame:** We put the trees into a dataframe that has as columns: the percent of full genome (0 through 100), the partition (as produced by prop.part), the bootstrap value of that partition (NA if doesn't exist in tree for percent of full genome, otherwise bootstrap value). For code, see actual notebook file.

We keep the labels separate from the dataframe for now because the names are extremely messy to keep as rows.

Figure 1: How similar are the trees?

```
library(phangorn)
```

Attaching package: 'phangorn'

The following object is masked from 'package:aricode':

RI

The following object is masked from 'package:plotly':

add_boxplot

```
dists <- lapply(sampletrees, function(x) {  
  just_trees <- lapply(x[[1]], function(y) y@phylo)[c(11,1,2,3,4,5,6,7,8,9,10)]  
  class(just_trees) <- "multiPhylo"  
  tree_rfdist <- dist.topo(just_trees)  
  #tree_rfdist <- RobinsonFoulds(just_trees, normalize=TRUE)  
  #tree_rfdist <- RF.dist(just_trees, normalize=TRUE, check.labels=FALSE)  
  melt(as.matrix(tree_rfdist), varnames = c("row", "col"))  
})  
df_dists <- bind_rows(dists, .id="id")  
#tiff(filename=file.path(PATH,"figures/fig1.tif"),res=300,width=1600,height=1200)  
ggplot(filter(df_dists, row=="wgs100" & value != 0), aes(x=col, y=value)) + geom_boxplot()
```

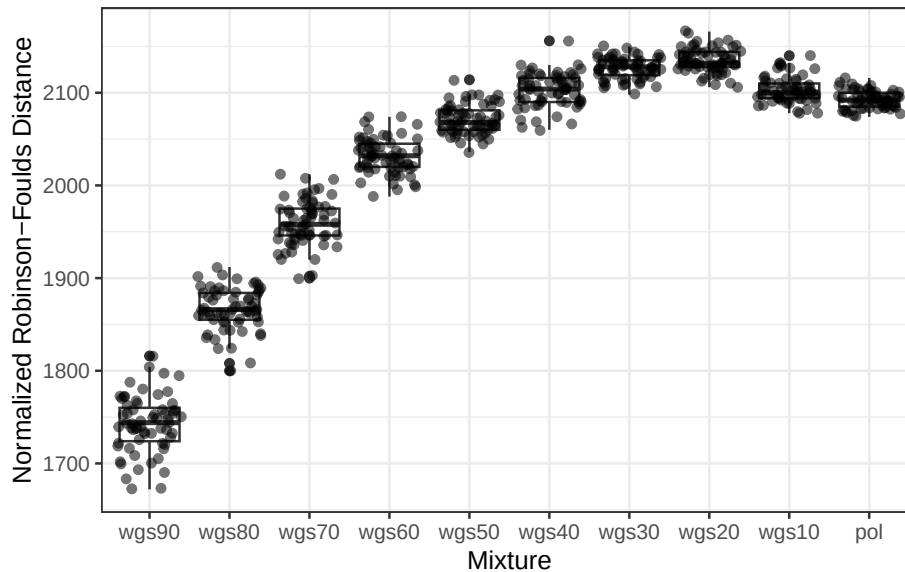


Figure 1: RF distances between all phylogenetic tree mixtures and the wgs tree. Closer to 0 on the y-axis is closer to the wgs tree, further is further away.

```
#dev.off()
```

We compute the RF distance between all of our phylogenetic tree samples and create a boxplot for them (Figure @ref{fig:fig1_rfdist}). It does appear that the pol tree is most similar to the pol10-wgs90 tree, and that the RF distance increases as the wgs proportion increases.

```
wgs50_dist <- filter(df_dists, row=="wgs100" & col=="wgs50")$value
pol_dist <- filter(df_dists, row=="wgs100" & col=="pol")$value
wgs40_dist <- filter(df_dists, row=="wgs100" & col=="wgs40")$value
wgs30_dist <- filter(df_dists, row=="wgs100" & col=="wgs30")$value
t.test(wgs50_dist)
```

One Sample t-test

```
data: wgs50_dist
t = 1050.1, df = 62, p-value < 2.2e-16
alternative hypothesis: true mean is not equal to 0
95 percent confidence interval:
 2067.042 2074.926
sample estimates:
```

```
mean of x  
2070.984
```

```
t.test(wgs50_dist, pol_dist, paired=TRUE, alternative="less")
```

```
Paired t-test
```

```
data: wgs50_dist and pol_dist  
t = -9.5214, df = 62, p-value = 4.791e-14  
alternative hypothesis: true mean difference is less than 0  
95 percent confidence interval:  
-Inf -17.98471  
sample estimates:  
mean difference  
-21.80952
```

Figure 2: PCoA of tree distances

```
#samplephylos <- lapply(sample.int(length(sampletrees), 20), function(x) lapply(sampletrees, x))
#distances <- ClusteringInfoDistance(samplephylos)
#write_rds(file=file.path(RDATA,"clustering_sample_distances.rds"),distances)
distances <- read_rds(file=file.path(RDATA, "clustering_sample_distances.rds"))

test_dist <- as.matrix(distances)[-c(11,22,33,44,55,66,77,88,99,110,121,132,143,154,165,176)]
pcoa <- cmdscale(test_dist, eig=TRUE, k=12)
mapping <- pcoa$points[,1:2]
#mapping <- pcoa$points[,2:3]
mix <- rownames(mapping)
mapping <- data.frame(mapping, Mixture=mix)

tiff(filename=file.path(PATH,"figures/fig2.tif"),res=300,width=1600,height=1600)
ggplot(mapping) + geom_point(aes(x=X1, y=X2, color=Mixture)) + theme_bw() + coord_fixed(xlim=0.0,ylim=0.0) +
  labs(color="Mixture") + xlab("X (8.09% Percent Variance)") + ylab("Y (2.59% Percent Variance)") +
  dev.off()
```

pdf
2

Figure @ref{fig:fig2_pcoa}. To get the percent variance, we took the eigenvalues of the pcoa object and divided the first 2 by the total sum:

- `pcoa$eig[1]/sum(pcoa$eig)`: 0.0808691
- `pcoa$eig[2]/sum(pcoa$eig)`: 0.0258566

Clustering results

Figure 3: Clustering changes with mixtures and bootstraps

```
# number of total clusters
samples_totalclust <- lapply(samples_allclusters, function(threshold) {
  tcdf <- sapply(threshold, function(y) sapply(y, function(x) length(x[[3]]))) %>%
    data.frame()
  names(tcdf) <- c("70", "80", "85", "90", "95", "99")
  tcdf$tree <- rownames(tcdf)
  tcdf <- pivot_longer(tcdf, cols=c("70", "80", "85", "90", "95", "99"), names_to="bootstrap")
  return(tcdf)
})

samples_totalclust <- bind_rows(samples_totalclust, .id="id")
samples_totalclust$tree[samples_totalclust$tree=="wgs"] <- "wgs100"
samples_totalclust$tree <- factor(samples_totalclust$tree, levels=c("pol", "wgs10", "wgs20", "wgs100"))
samples_totalclust$numerical_tree <- c(0,10,20,30,40,50,60,70,80,90,100)[match(samples_totalclust$tree, levels)]

# size of clusters
samples_clustsize <- lapply(samples_allclusters, function(sample) {
  df <- lapply(sample, function(threshold) {
    sizes <- sapply(threshold, function(y) sapply(y[[1]], length))
    sizes_df <- bind_rows(lapply(sizes, data.frame), .id="tree")
    colnames(sizes_df) <- c("tree", "size")
    return(sizes_df)
  })
  df <- bind_rows(df, .id="threshold")
  return(df)
})

samples_clustsize <- bind_rows(samples_clustsize, .id="id")
samples_clustsize$tree[samples_clustsize$tree=="wgs"] <- "wgs100"
samples_clustsize$tree <- factor(samples_clustsize$tree, levels=c("pol", "wgs10", "wgs20", "wgs100"))
```

```
colnames(samples_clustsize) <- c("id", "bootstrap", "tree", "size")
samples_clustsize$numerical_tree <- c(0,10,20,30,40,50,60,70,80,90,100)[match(samples_clu
```

For the number of clusters and size of clusters, we compute two dataframes, `samples_totalclust` and `samples_clustsize`. The `samples_totalclust` will be used later as well to help compute AMI, but for `sample_clustsize` we will compute the mean cluster size to summarize the cluster size distribution in a dataframe `samples_mean_clustsize`. In general all of them follow a Poisson distribution with mostly clusters of size 2, with larger clusters as more rare. (Supplementary Figure)

One sample, 15 was left out as it was a major outlier in cluster size:
`filter(samples_clustsize, id==15) %>% group_by(bootstrap, tree)`
`%>% summarise(max=max(size)).`

For the cluster size and number of clusters, we make both a table and a side-by-side plot of cluster sizes and numbers.

```
samples_totalclust <- filter(samples_totalclust, id != 15)
samples_clustsize <- filter(samples_clustsize, id != 15)

samples_mean_clustsize <- samples_clustsize %>%
  group_by(id, bootstrap, tree) %>%
  summarise(mean=mean(size))
```

``summarise()`` has grouped output by 'id', 'bootstrap'. You can override using the ``.groups`` argument.

```
filter(samples_totalclust, id != 15) %>% group_by(bootstrap, tree) %>% summarise(mean=mean
```

``summarise()`` has grouped output by 'bootstrap'. You can override using the ``.groups`` argument.

```
# A tibble: 6 x 12
# Groups:   bootstrap [6]
  bootstrap    pol wgs10 wgs20 wgs30 wgs40 wgs50 wgs60 wgs70 wgs80 wgs90 wgs100
  <chr>      <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 70        258. 251. 248. 244. 251. 247. 247. 246. 248. 248. 240
2 80        256. 251. 243. 241. 243. 241. 239. 240. 243. 245. 241
3 85        251. 245. 234. 232. 234. 234. 230. 231. 235. 240. 245
4 90        238. 231. 220. 218. 217. 217. 214 214. 222. 226. 241
5 95        209. 205 195. 189. 188. 188. 187. 188. 198. 204. 216
6 99        149. 147 144. 139. 138. 136 140. 145. 155. 165. 176
```

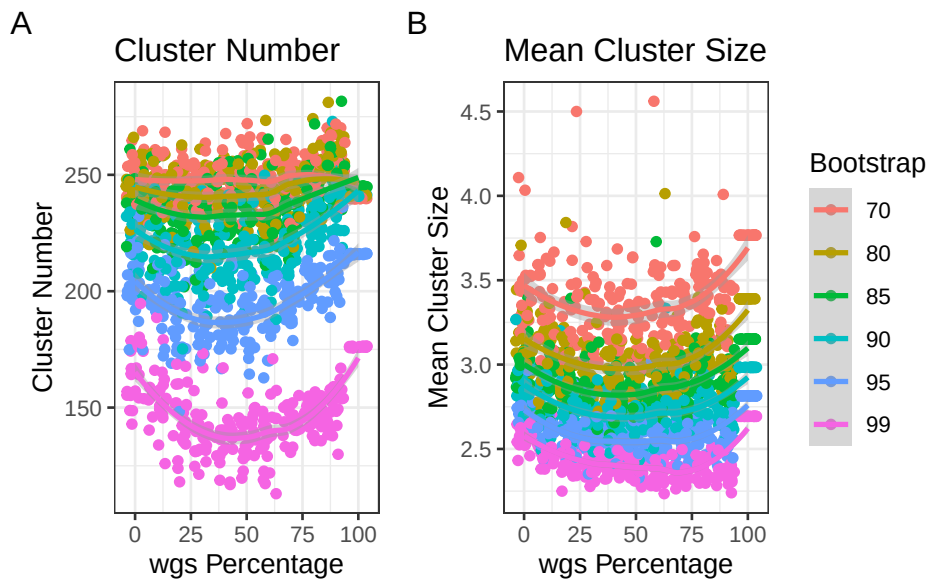
```
samples_clustchange <- left_join(samples_totalclust, samples_mean_clustsize)
```

Joining with `by = join\_by(id, tree, bootstrap)`

```
library(patchwork)
p1 <- ggplot(samples_clustchange, aes(x=numerical_tree,y=totalclust)) + geom_jitter(aes(c
  theme_bw() +
  xlab("wgs Percentage") +
  ylab("Cluster Number") +
  theme(legend.position="none") +
  ggtitle("Cluster Number")
p2 <- ggplot(samples_clustchange, aes(x=numerical_tree,y=mean,color=bootstrap)) + geom_ji
  theme_bw() +
  xlab("wgs Percentage") +
  ylab("Mean Cluster Size") +
  ggtitle("Mean Cluster Size") +
  guides(color=guide_legend(title="Bootstrap"))
#tiff(filename=file.path(PATH,"figures/fig3.tif"),res=300,width=1800,height=1400)
p1 + p2 + plot_annotation(tag_levels = 'A')
```

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'

`geom\_smooth()` using method = 'loess' and formula = 'y ~ x'



```
#dev.off()
```

```
table_clustchange <- samples_clustchange %>% group_by(bootstrap, tree) %>%
  summarise("Mean Number of Clusters"=mean(totalclust), "Mean Cluster Size"=mean(mean))
  pivot_longer(names_to="stat", cols=c("Mean Number of Clusters","Mean Cluster Size"))
  pivot_wider(names_from=tree, values_from=value)
```

`summarise()` has grouped output by 'bootstrap'. You can override using the
 `.groups` argument.

```
library(flextable)
```

Warning: package 'flextable' was built under R version 4.3.3

Attaching package: 'flextable'

The following objects are masked from 'package:kableExtra':

```
as_image, footnote
```

The following objects are masked from 'package:plotly':

```
highlight, style
```

The following object is masked from 'package:purrr':

```
compose
```

The following object is masked from 'package:ape':

```
rotate
```

The following object is masked from 'package:ggtree':

```
rotate
```

```
ft1 <- samples_clustchange %>%
  filter(id != 15) %>%
  group_by(bootstrap, tree) %>%
  summarise("Number"=mean(totalclust), "Size"=mean(mean)) %>%
  pivot_longer(names_to="stat", cols=c("Number","Size")) %>%
  pivot_wider(names_from=c(bootstrap,stat), values_from=value) %>%
  flextable() |>
  colformat_double() |>
  theme_vanilla() |>
```

Table 1: Mean number of clusters and cluster size at different bootstrap thresholds and mixtures. Each value is the mean over the samples run.

Stat	pol	wgs10	wgs20	wgs30	wgs40	wgs50	wgs60	wgs100
Bootstrap=70								
Mean Number of Clusters	257.826	250.957	248.217	244.217	250.870	247.435	247.130	245.900
Mean Cluster Size	3.382	3.457	3.319	3.357	3.270	3.294	3.281	3.300
Bootstrap=80								
Mean Number of Clusters	256.391	251.130	243.087	240.957	242.652	241.348	239.043	240.200
Mean Cluster Size	3.045	3.112	3.010	3.010	2.968	2.979	2.999	3.000
Bootstrap=85								
Mean Number of Clusters	251.043	244.522	234.304	232.478	234.087	234.348	229.913	231.200
Mean Cluster Size	2.888	2.956	2.872	2.869	2.816	2.806	2.856	2.800
Bootstrap=90								
Mean Number of Clusters	238.391	230.783	219.609	217.957	216.826	216.696	214.000	213.800
Mean Cluster Size	2.710	2.783	2.728	2.714	2.684	2.682	2.721	2.700
Bootstrap=95								
Mean Number of Clusters	208.522	205.000	194.957	189.478	188.087	187.565	187.043	187.600
Mean Cluster Size	2.552	2.588	2.555	2.548	2.551	2.539	2.558	2.500
Bootstrap=99								
Mean Number of Clusters	149.435	147.000	143.870	138.652	138.174	136.000	139.652	145.100
Mean Cluster Size	2.358	2.398	2.376	2.370	2.404	2.401	2.418	2.400

```

separate_header() %>%
add_header_row(values=c("Mixture","Bootstrap"),
               colwidths=c(1,12)) %>%
labelizor(part="header",labels=c("tree="")) %>%
add_footer_lines("Mean number of clusters and cluster size at different bootstrap thresholds")
set_caption(caption="Cluster statistics by mixture and threshold.")

```

`summarise()` has grouped output by 'bootstrap'. You can override using the `.groups` argument.

```

save_as_docx(ft1,path="cluster_table.docx")

kbl(select(ungroup(table_clustchange),stat:wgs100), digits=3, col.names=c("Stat","pol","wgs"))

# kbl(select(ungroup(table_clustchange),stat:wgs100), digits=3, col.names=c("Stat","pol","wgs"))

ggplot(samples_clustsize, aes(x=size,fill=tree)) + geom_bar(position="dodge") + facet_wrap(~bootstrap)

```

Warning: Removed 83 rows containing non-finite outside the scale range
(`stat\_count()`).

Warning: Removed 5 rows containing missing values or values outside the scale range
(`geom\_bar()`).

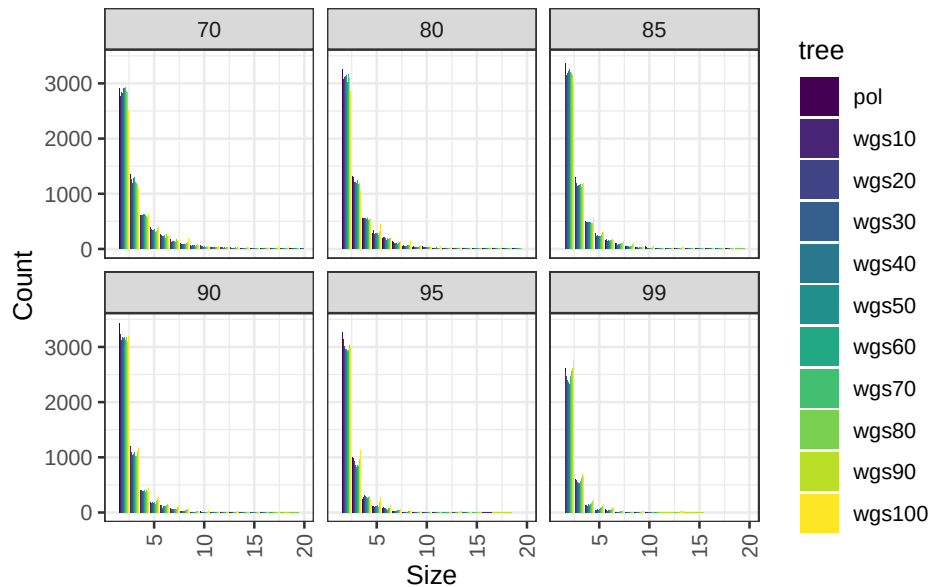


Figure 2: Supplementary figure of cluster size

Sizes of clusters would be from `samples_allclusters`. Can provide average cluster size How change with changing mixtures and bootstraps - table
`filter(samples_clustsize, size>200)`

```
test7 <- filter(samples_clustsize, id != 15) %>% group_by(threshold, tree) %>% summarise(

test8 <- filter(samples_clustsize, id != 15) %>% group_by(id, threshold, tree) %>% summarise(
test8$tree <- c(0,10,20,30,40,50,60,70,80,90,100)[match(test8$tree, treenames)]
ggplot(test8, aes(x=tree,y=mean)) + facet_wrap(~threshold) + geom_jitter(aes(color=tree))
p2 <- ggplot(test8, aes(x=tree,y=mean)) + geom_jitter(aes(color=threshold)) + geom_smooth

#ggplot(filter(samples_clustsize, id != 15), aes(x=tree,y=size,fill=tree)) + geom_violin(

#test7 <- filter(samples_clustsize, id != 15) %>% group_by(threshold, tree) %>% summarise(

# test7 <- samples_clustsize
# test7$threshold <- as.numeric(test7$threshold)
# test7$tree <- c(0,10,20,30,40,50,60,70,80,90,100)[match(samples_clustsize$tree, treenames)]
```

```

threshold <- c(70, 80, 85,90,95,99)

# number of common clusters = tp
samples_sharedclust <- lapply(samples_allclusters, function(threshold) {
  tdf <- sapply(threshold, function(y) sapply(y, function(x) intersect(x[[3]],y$wgs[[3]]))
  data.frame()
  names(tdf) <- c("70","80","85","90","95","99")
  tdf$tree <- rownames(tdf)
  tdf <- pivot_longer(tdf, cols=c("70","80","85","90","95","99"), names_to="bootstrap",
  return(tdf)
})
samples_sharedclust <- bind_rows(samples_sharedclust, .id="id")

# number of total - common = clusters specific to given tree = fp
# number of wgs - common = clusters specific to wgs = fn
# https://stackoverflow.com/questions/50957998/mutate-inside-group-by-based-on-particular
# number of possible clusters is number of nodes in tree which here we know is 1194
# 1194 - fn = tn
samples_cluster_compare <- samples_totalclust %>% left_join(samples_sharedclust) %>%
  mutate(fp=totalclust-tp) %>%
  group_by(bootstrap) %>%
  mutate(fn=totalclust[tree=="wgs"]-tp, tn=1194-fn, precision=tp/(tp+fp),recall=tp/(tp+fn))
samples_cluster_compare$tree <- factor(samples_cluster_compare$tree, order=TRUE, levels=c("wgs","tree"))
samples_cluster_compare <- mutate(samples_cluster_compare, F1=(2*precision*recall)/(precision+recall))

ggplot(filter(samples_cluster_compare, tree!="wgs"),aes(y=precision,x=recall,color=tree))

```

Figure 4: Clustering AMIs for trees

```

samples_clustid <- lapply(samples_allclusters, function(sample) {
  tdf <- lapply(sample, function(threshold) {
    lapply(seq_along(threshold), function(n) {
      mix <- threshold[[n]]
      mixname <- names(threshold)[n]
      mixid <- mix[[2]]
      names(mixid) <- 1:length(mixid)
      mixdf <- unlist(mixid)
      mixdf <- data.frame(id=mixdf, clust=names(mixdf)) %>%
        mutate(clust=substr(clust,1,nchar(clust)-1))
      names(mixdf) <- c("id",mixname)
      return(mixdf)
    }) %>%
    reduce(full_join,by=c("id")) %>%
    replace(is.na(.),0) # replace all NAs
  })
  names(tdf) <- c("70","80","85","90","95","99")
  tdf <- bind_rows(tdf,.id="threshold")
  return(tdf)
})

samples_AMI <- lapply(samples_clustid, function(sample) {
  sample %>% group_by(threshold) %>% summarise(across(wgs90:pol, ~AMI(.x,wgs))) %>%
    pivot_longer(cols=wgs90:pol, names_to="mix", values_to="AMI")
})

samples_AMImean <- bind_rows(samples_AMI,.id="id") %>% group_by(threshold,mix) %>% summarise(

# test4 <- bind_rows(samples_clustid[[1]], samples_clustid[[2]], samples_clustid[[3]], sa
#
# group_by(threshold) %>% summarise(across(wgs90:wgs, list(wgs=~AMI(.x,wgs),wgs90=~AMI(.x

```



```

# wgs70
# wgs40
# wgs10
# pivot_longer(cols=wgs90_wgs:wgs_pol,names_to="mix_mix",values_to="AMI") %>%
# separate(col="mix_mix",into=c("mix1","mix2")) -> test
# test2 <- unite(test, "mix11", mix1, id, remove=FALSE) %>% unite("mix21", mix2, id)
# autoplot(cmdscale(1-acast(filter(test,threshold==99),mix1-mix2,value.vaar="AMI"),ei
#
# testmat <- left_join(filter(samples_clustid[[1]], threshold==99), filter(samples_cl
#
# testami <- lapply(3:35, function(x) lapply(3:35, function(y) AMI(testmat[,x],testma
# testami_mat <- acast(melt(testami),L2~L1)
# ggplot(data.frame(cmdscale(1 - testami_mat),mix=c(treenames,treenames,treenames)))
#
# """"
#

ggplot(bind_rows(samples_AMI,.id="id") %>% filter(id != 15),aes(y=AMI,x=mix)) + geom_boxp
theme_bw() +
theme(axis.text.x=element_text(angle=90,vjust=0.5,hjust=1))

samples_clustid_wgsonly <- lapply(samples_clustid, function(x) filter(x, wgs != 0))
samples_AMI_wgsonly <- lapply(samples_clustid_wgsonly, function(sample) {
  sample %>% group_by(threshold) %>% summarise(across(wgs90:pol, ~AMI(.x,wgs))) %>%
    pivot_longer(cols=wgs90:pol, names_to="mix", values_to="AMI")
})
samples_AMImean_wgsonly <- bind_rows(samples_AMI_wgsonly,.id="id") %>% group_by(threshold

```

Parameters for trees

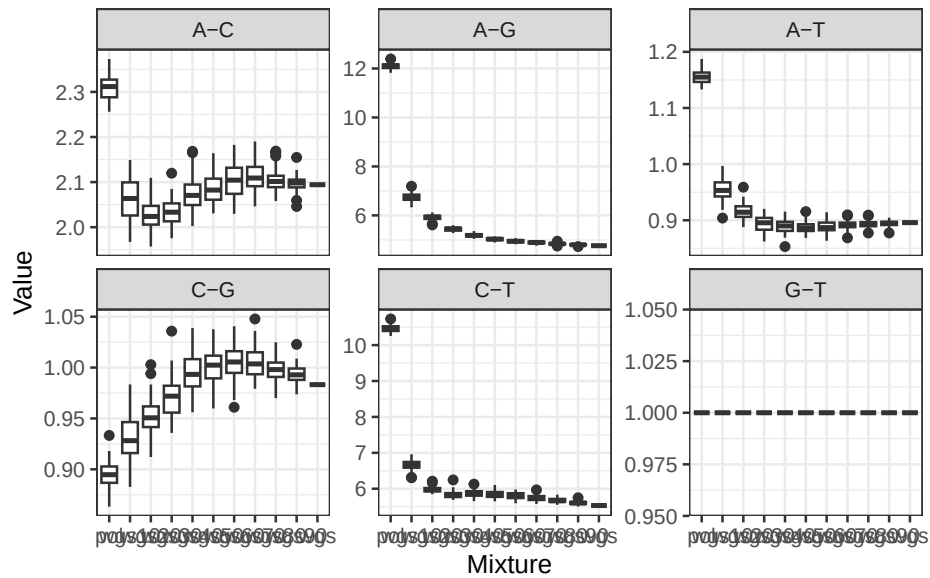
```

wgsparms <- paramsdf(file.path(PATH,"results/trees/HIV1_FLT_2018_genome_DNA_subtypeB.fa.
folders <- list.files(file.path(PATH,"results/trees/samples"))
sampleparams <- lapply(folders, function(x) {
  loadparams(file.path(PATH,"results/trees/samples",x))
})
sampleparams <- Filter(Negate(is.null), sampleparams)

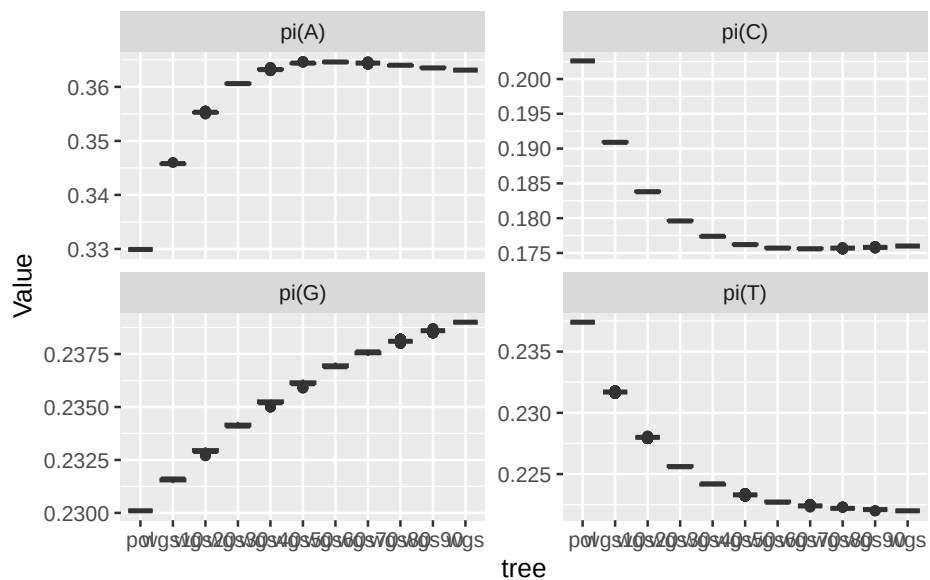
paramdf <- lapply(sampleparams, function(x) bind_rows(x, .id="tree")) %>% bind_rows(., .id
wgsparms$id <- 1000
wgsparms$tree <- "wgs"

```

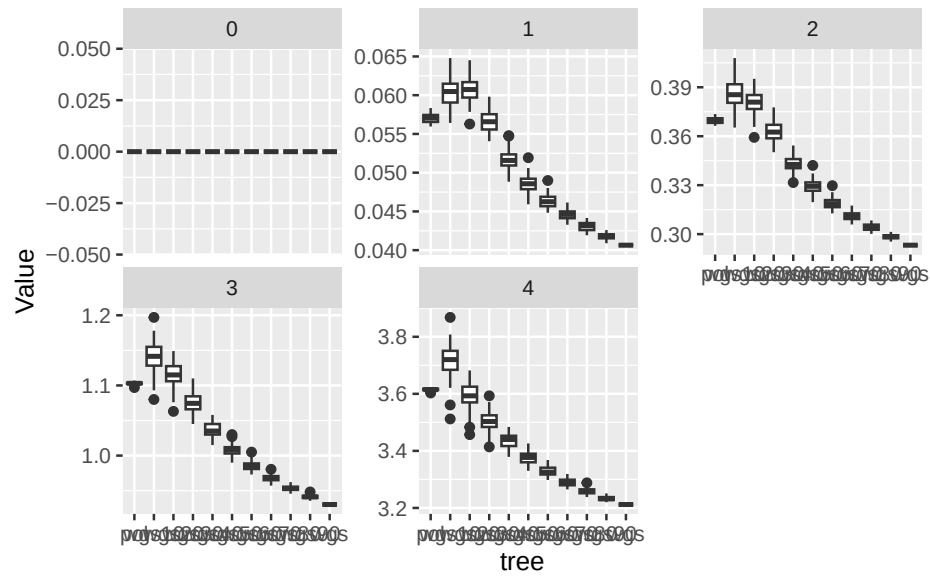
```
paramdf <- rbind(paramdf, wgsparams) %>%
  mutate(tree=fct_relevel(tree,"pol","wgs10","wgs20","wgs30","wgs40","wgs50","wgs60","wgs70","wgs80","wgs90","wgs100"))
ggplot(filter(paramdf, Type=="Rate parameter R"), aes(x=tree,y=Value)) + geom_boxplot() +
```



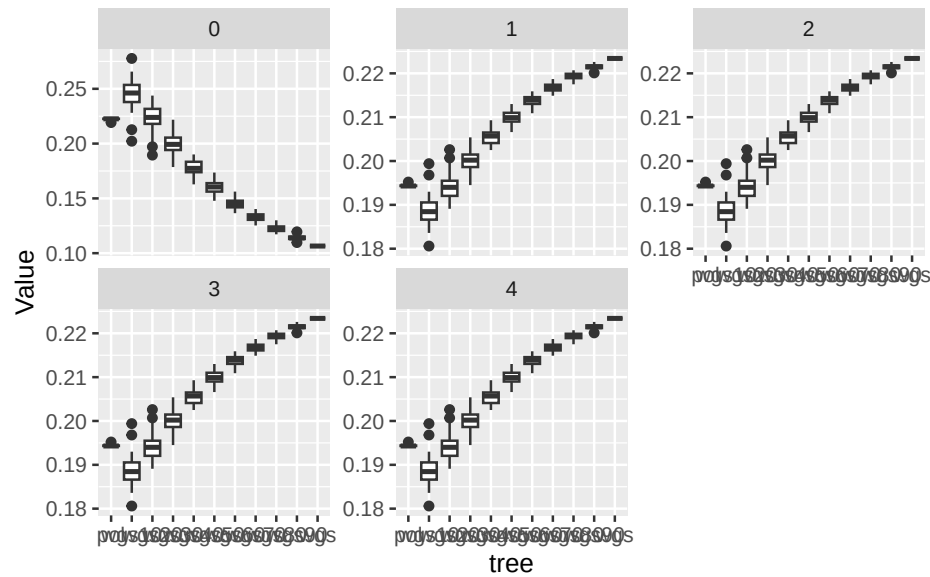
```
ggplot(filter(paramdf, Type=="State freq"), aes(x=tree,y=Value)) + geom_boxplot() + facet
```



```
ggplot(filter(paramdf, Type=="relative rate"), aes(x=tree,y=Value)) + geom_boxplot() + fa
```



```
ggplot(filter(paramdf, Type=="rr proportion"), aes(x=tree,y=Value)) + geom_boxplot() + fa
```



```

fpdists <- lapply(fptrees, function(x) {
  just_trees <- lapply(x[[1]], function(y) y@phylo)[c(11,1,2,3,4,5,6,7,8,9,10)]
  just_trees <- lapply(just_trees, function(y) {
    y$node.label <- NULL
    return(y)
  })
  # there is also one from 1979 in here?
  # sapply(just_trees$wgs$tip.label, function(x) stringr::str_split(x, pattern="\\.")[[
  # which(sapply(just_trees$wgs$tip.label, function(x) stringr::str_split(x, pattern="\\
  # B.US.79.NYC4.KJ704792
  #
  # 296
  #just_trees <- lapply(just_trees, function(x) ape::root(x,outgroup="B.FR.83.HXB2_LAI_
  #just_trees <- lapply(just_trees, function(x) ape::root(x,outgroup="B.US.79.NYC4.KJ70
  class(just_trees) <- "multiPhylo"
  tree_rfdist <- dist.topo(just_trees)
  #tree_geodist <- dist.multiPhylo(just_trees)
  melt(as.matrix(tree_rfdist), varnames = c("row", "col"))
  #melt(as.matrix(tree_geodist), varnames = c("row", "col"))
})
df_fpdists <- bind_rows(fpdists, .id="id")
ggplot(filter(df_fpdists, row=="wgs" & value != 0), aes(x=col, y=value)) + geom_boxplot()

```



Figure 3: Sample to sample RF distances between all phylogenetic trees. Closer to yellow is farther away, and closer to green is closer.

```

lapply(1:4, function(x) {
  st <- sampletrees[[x]]
  st <- lapply(st[[1]], function(y) y@phylo)
  ft <- fptrees[[x]]
  ft <- lapply(ft[[1]], function(y) y@phylo)
  d <- mapply(dist.multiPhylo, st, ft)
})

```

How many clusters are identified in each mixture compared to whole genome?

```

threshold <- c(70, 80, 85,90,95,99)
allclusters <- lapply(threshold, function(y) future_lapply(trees, function(x) cluster(x,
names(allclusters) <- threshold

# number of common clusters = tp
sharedclust <- lapply(allclusters, function(y) sapply(y, function(x) intersect(x[[3]],y$w
tp <- data.frame(sharedclust)
names(tp) <- c("70","80","85","90","95","99")
tp$tree <- rownames(tp)
tp <- pivot_longer(tp, cols=c("70","80","85","90","95","99"), names_to="bootstrap", value
# number of total clusters
totalclust <- lapply(allclusters, function(y) sapply(y, function(x) length(x[[3]])))

# number of total - common = clusters specific to given tree = fp
fp <- data.frame(totalclust)
names(fp) <- c("70","80","85","90","95","99")
fp$tree <- rownames(fp)

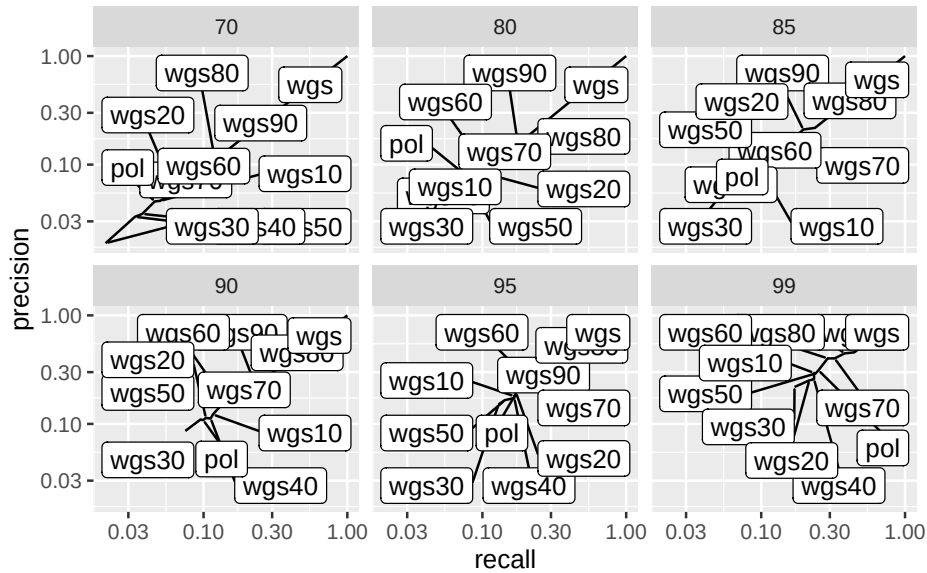
# number of wgs - common = clusters specific to wgs = fn
# https://stackoverflow.com/questions/50957998/mutate-inside-group-by-based-on-particular

# number of possible clusters is number of nodes in tree which here we know is 1194
# 1194 - fn = tn
cluster_compare <- pivot_longer(fp, cols=c("70","80","85","90","95","99"), names_to="boot
mutate(fp=totalclust-tp) %>%
group_by(bootstrap) %>%
mutate(fn=totalclust[tree=="wgs"]-tp, tn=1194-fn, precision=tp/(tp+fp),recall=tp/(tp+

```

Joining with `by = join\_by(tree, bootstrap)`

```
ggplot(cluster_compare,aes(y=precision,x=recall)) + geom_line() + ggrepel::geom_label_repel
```



```
sizeclust <- lapply(allclusters, function(x) lapply(x, function(y) sapply(y[[1]], length)
```

```
threshold <- c(70, 80, 85,90,95,99)
maskgp120_allclusters <- lapply(threshold, function(y) future_lapply(maskgp120_phylos, fu
names(allclusters) <- threshold

# number of common clusters = tp
sharedclust <- lapply(maskgp120_allclusters, function(y) sapply(y, function(x) intersect(
tp <- data.frame(sharedclust)
names(tp) <- c("70","80","85","90","95","99")
tp$tree <- rownames(tp)
tp <- pivot_longer(tp, cols=c("70","80","85","90","95","99"), names_to="bootstrap", value
# number of total clusters
totalclust <- lapply(maskgp120_allclusters, function(y) sapply(y, function(x) length(x)[3

# number of total - common = clusters specific to given tree = fp
fp <- data.frame(totalclust)
names(fp) <- c("70","80","85","90","95","99")
fp$tree <- rownames(fp)

# number of wgs - common = clusters specific to wgs = fn
```

```

# https://stackoverflow.com/questions/50957998/mutate-inside-group-by-based-on-particular

# number of possible clusters is number of nodes in tree which here we know is 1194
# 1194 - fn = tn
cluster_compare <- pivot_longer(fp, cols=c("70","80","85","90","95","99"), names_to="bootstrap",
  mutate(fp=totalclust-tp) %>%
  group_by(bootstrap) %>%
  mutate(fn=totalclust[tree=="wgs"]-tp, tn=1194-fn, precision=tp/(tp+fp), recall=tp/(tp+fn))

ggplot(cluster_compare, aes(y=precision, x=recall)) + geom_line() + ggrepel::geom_label_repel()

```